

Team alignment — what we are building and why

The hackathon we are entering

C:>DIR Global "Agentic Regulator" Hackathon, Problem Statement 4: Know Your Agent (KYA). The brief asks teams to build something that authenticates and verifies AI agents, and detects when they are doing things they are not supposed to do. It lists four challenge pathways teams can focus on. We are focusing on one.

Our chosen challenge pathway: Delegation Gap

The brief defines this as: *"Verifying agents stay within authorized scope."*

In plain English — when a human tells an AI agent to do something on their behalf (like make a payment), how does a bank or institution know the agent is genuinely acting within what the human actually approved? Right now, it can't. That is the gap. We are closing it.

Our product: Agent Passport

Agent Passport is a credential system for AI agents making financial transactions. Here is what it does step by step:

1. A human customer approves a payment mandate — specifying who the money can go to, how much, and for what purpose. This is signed by the customer cryptographically (using Ed25519, a standard digital signature method).
2. A regulated AI provider issues an identity credential for the agent — proving the agent is from a certified, regulated provider. This is also cryptographically signed.
3. A regulatory authority signs off that the provider meets the required standards.

These three signatures together form what we call the **Agent Passport** — a tamper-proof credential envelope the agent carries when it submits a payment.

When the payment hits the bank, the bank runs a deterministic check (no AI, no guesswork — pure cryptographic verification) in under 10 milliseconds. It checks: is the agent's identity valid? Is the provider regulated? Does the action fall within what the human actually approved? If any check fails, the payment is denied. If the agent has been manipulated — for example, a poisoned invoice redirecting money to a different account — the mandate does not cover that action, so it is caught and blocked.

The two components inside Agent Passport

The hackathon brief lists illustrative prototype concepts. Agent Passport contains two of them, and they work together as one system:

Component 1 — Standards Review Assistant (what the brief calls a "KYA standards evaluation tool")

This is a regulatory copilot used *before* a passport is issued. It helps compliance officers check whether an AI provider meets the required regulatory standards — NIST, FATF, PSR 2017, etc. The tool surfaces which standards apply, flags gaps, and produces a signed audit record. A human makes every decision. The tool assists. This is the front door: no provider gets to issue Agent Passports unless they pass this review.

Component 2 — Cryptographic Delegation Verifier (what the brief calls a "cryptographic authorization validator")

This is what runs *at the moment of every payment*. When an agent submits a transaction to a bank, this verifier checks the full chain: (a) is the agent's cryptographic identity valid, (b) is the delegation from the customer's mandate to this agent's action valid, and (c) does the action fall strictly within the scope the human approved? The key rule is: delegation can only narrow authority, never expand it. An agent cannot grant itself more permissions than the human gave it. This is enforced mathematically, not by policy. If the chain breaks at any point, the bank denies the payment.

Why these two components belong together

The Standards Review Assistant ensures only trustworthy providers can issue Agent Passports in the first place. The Cryptographic Delegation Verifier ensures that even a passport from a trustworthy provider cannot be used to exceed what the human actually approved. One handles entry to the system; the other handles every transaction within it. You need both. A credential from an unvetted provider is worthless. A vetted provider whose agents can still be manipulated into acting outside mandate is also worthless.

What we are NOT doing

We are not building age verification, cross-border trust layers, or DPI risk simulation — the other prototype concepts in the brief. We are not covering multiple challenge pathways. We are solving one pathway (Delegation Gap) completely, with a working demo. That is the submission.

The demo in three minutes

The demo shows this sequence: a valid agent makes a clean payment — it passes. The same valid agent is then manipulated by a poisoned invoice trying to redirect funds to a different account. The mandate does not cover that beneficiary. The bank's verifier catches it. The payment is denied. A supervisor sees the violation, reviews it, and revokes the passport. The audit trail is signed and complete. That is the whole story in three minutes.